

Lab 10 - Rotational Equilibrium (CH9.1)

GROUP #8, Jared Lomeli, Justas Radavicius, Ethan Dickson-Child, Juanita Taborda Arias

Rationale This lab will be done in advance of CH9 Statics and Torque. You will have the opportunity to explore an extension of Newton's Laws to a system that is free to rotate.

Learning Objectives

- Collaborate with other students in problem-solving and laboratory activities.
- Analyze real-world experimental data
- Apply a problem-solving strategy
- Display good scientific practices

Equipment/Apparatus

- meter stick ("pole")
- four balancing clamps
- mass scale or balance
- various masses, as needed
- personal laptop (no capstone)

Tips to meet or exceed expectations:

- **Measurements** should have error bars and the correct number of significant figures
- **Extended free-body diagrams** should represent the general shape of the objects, and the forces applied on the objects should be drawn **where they are acting**.
- **Use the steps of the problem-solving method:**
 - Draw a representation of the problem (sketch of the situation)
 - Indicate known and unknown values on the representation
 - List useful equations
 - Solve for the unknown
 - Assess if the answers make physical sense

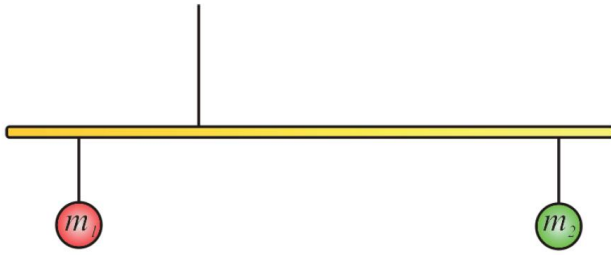
Prelab

1. Brainstorm ideas about the following terms based on your experience, then record the definitions using the College Physics OpenStax textbook or your favorite search engine/AI:

Term	Brainstorm	Definition
Torque	Movement of an object in the direction of force	The turning or twisting effectiveness of a force The measure of the rotational force acting on an object, forcing rotation along the axis at location "r" (Vector = Magnitude and direction) $\tau = rF \sin \theta$
Center of mass	The center of the object with mass The point where the object is in balance	The point where the total mass of the body is assumed to be concentrated
Center of gravity	The balance point of an object Similar to the center of mass?	"the point where the total weight of the body is assumed to be concentrated"
Axis of rotation	The point a subject rotates around A mix of x and y axis to make up the axis in which it is rotating?	A fixed point in which other points moved around that fixed point
Static equilibrium	The equilibrium of an object that is non-moving A state where the object is balanced	"a state of equilibrium in which the net external force and torque acting on a system is zero"
Extended free-body diagram	A free body diagram that shows interactions extending outside of just the subject	A tool used to analyze rotational motion and torque by representing an object as its actual shape rather than a single point

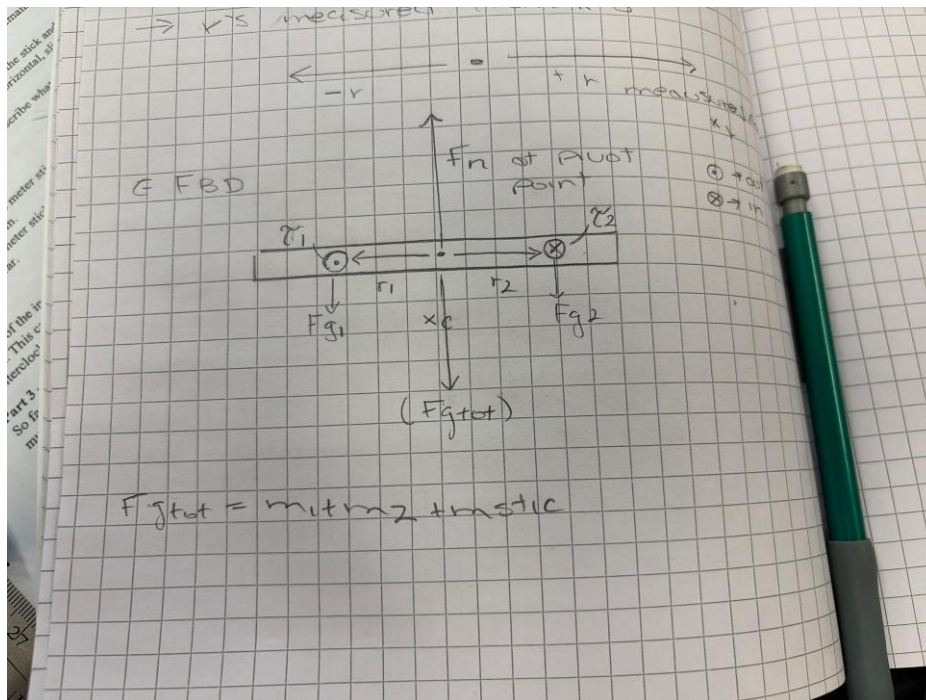
2. Compare answers and do example 9.1a in your notes as a class.

Stop and think (read and write)



A uniform meterstick of length L and mass m_m is to balance from a clamp (shown here as a line) at x_c . The center of mass of the pole is at x_m , where $x_m > x_c$, measured from $x=0$ at the left end of the pole. Mass m_1 is placed at a distance x_1 from the left end of the pole. Mass m_2 is placed at a distance x_2 from the left end of the pole.

1. Insert a sketch or drawing of the pole system as described. Label the masses, their positions and the forces at the positions where they are acting. This is an EFB!



2a. Write the equation for the condition of static equilibrium defined in 9.1. Does your group recognize it? Why or why not.

$net F = 0$ Yes, it is familiar because we have used it multiple times in class before. Usually when the $net F = 0$ That means that the displacement is zero and thus no work is done.

2b. Write the equation for the condition of static equilibrium defined in 9.2. Does your group recognize it? Why or why not.

$$\text{net } \tau = 0$$

3. Write the equation for torque in terms of each force F acting at a distance r from the axis of rotation. NOTE: r is NOT measured from the end of the pole! Add r_1 and r_2 to your sketch.

$$\tau = rF\sin\theta$$

4. Use torque equations and example 9.1 to determine the mass of m_1 placed at x_1 that will balance the pole at x_c

$$(x_c - x_1)(m_1g) - (x_m - x_c)(mg) - (x_2 - x_c)(m_2g) = 0$$

$$M_1 = \frac{m(x_m - x_c) + m_2(x_2 - x_c)}{(x_c - x_1)}$$

5. Express your answer in terms of g , x_m , m_m , x_1 , m_2 , x_c , and x_2 .

g is equal to the force of gravity earth = 9.81 m/s^2

x_m = the point where the mass is found

M_m = the mass of the meter stick

x_1 = is the distance from zero where the first mass is recorded

M_2 = the second mass being recorded

x_c = the center of mass

x_2 = the distance from zero where the second mass is being recorded

Predict the unknown mass

Measure the mass of the meter stick and record it as m_m .

$$m_m = 81.5g$$

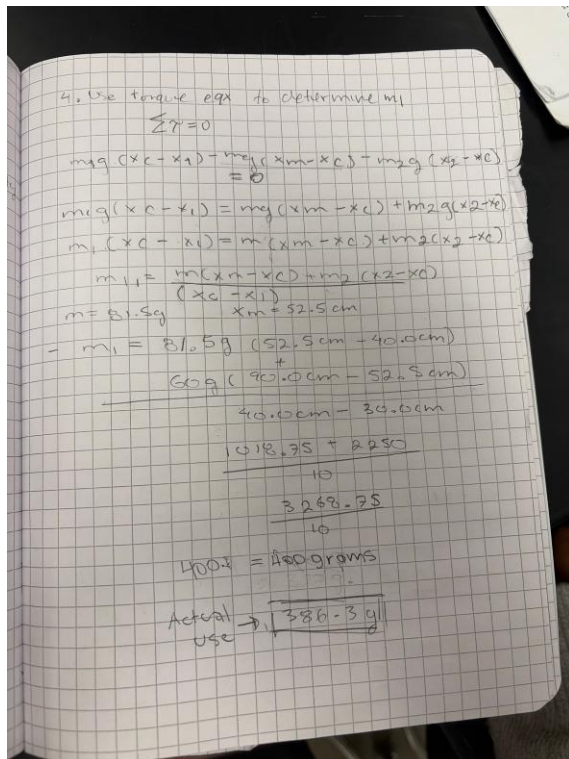
Locate the position of the center of mass of the meter stick relative to the 0 cm end of the stick and record this value as x_m . (Explain how you do this.)

$$52.5 \text{ cm} = x_m$$

We balanced the stick on a thin platform looking for the point where it would no longer tip to any side. When balanced mass is equal on both sides.

Using the equation you came up with in stop and think, determine the mass m_1 required for equilibrium if $x_c = 40.0 \text{ cm}$, $m_2 = 60 \text{ g}$, $x_1 = 30.0 \text{ cm}$, and $x_2 = 90.0 \text{ cm}$. Pro tip: You can use cm and g as units throughout this lab. The units will work out.

$$M_1 = 401.9 \text{ g}$$



Measure the mass of each clamp and incorporate this value into the total mass for each of the masses m_1 , m_2 , and m_3 , in the later experiment.

The masses of each of the clamps are 9.9 g

Attach a balancing clamp to the meter stick at the $x_c = 40.0$ cm mark. Suspend a total of 60 g (including the balancing clamp) from the 90.0 cm mark on the meter stick. Add mass to a second balancing clamp positioned at the 30.0 cm mark until the meter stick is level and in rotational equilibrium. **Once level, record the total mass of the clamp and added masses as m_1 .**

$$M_1 = 386.3 \text{ g}$$

Calculate the %-difference between the predicted m_1 and the experimental value.

Predicted = 401.9 g, actual = 386.3 g

$$\% \text{ error} = (401.9 - 386.3) / 401.9 \times 100\% = 3.88\%$$

How much farther away is mass 2 from the pivot point (x_c) than mass 1? Evaluate by calculating the ratio of r_2/r_1 . NOTE: Use the distance from the axis of rotation, NOT the end of the pole.

x_2 was 50 cm away from the pivot point.

x_1 was 10 cm away from the pivot point

$50/10 = r_2$ was 2 times bigger than r_1

How much more massive is mass 1 than mass 2? Evaluate by calculating the ratio m_1/m_2 .

$386.3\text{g} / 60\text{g} = \text{mass 1 is 6.44 times bigger than mass 2}$

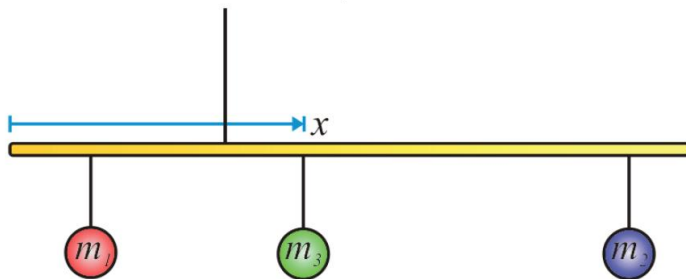
How does the ratio r_2/r_1 compare to m_1/m_2 ? Explain why they might be different.

$$r_2/r_1 = m_1/m_2$$

In theory

However, they are slightly different due to the center of mass of the ruler being at 52.5 cm rather than 50.0 cm

Stop and think again



A uniform pole of length L , mass m_m and center of mass at x_m is to balance x_c such that $x_m > x_c$. Masses m_1 and m_2 are attached to the pole at distances of x_1 and x_2 from the left end of the pole, respectively. If $x_1 < x_c < x_2$, **where should a third mass m_3 be placed such that the pole balances at x_c ? Express your answer in terms of m_m , m_1 , m_2 , m_3 , x_m , x_c , x_1 and x_2 . Show your work.**

$$M_1 = 100\text{g}$$

$$M_2 = 100\text{g}$$

$$M_3 = 200\text{g}$$

$$M_m = 81.5\text{g}$$

$$x_m = 1\text{m}$$

$$x_c = .0525\text{m}$$

$$x_1 = 35.0\text{cm}$$

$$x_2 = 95.0\text{cm}$$

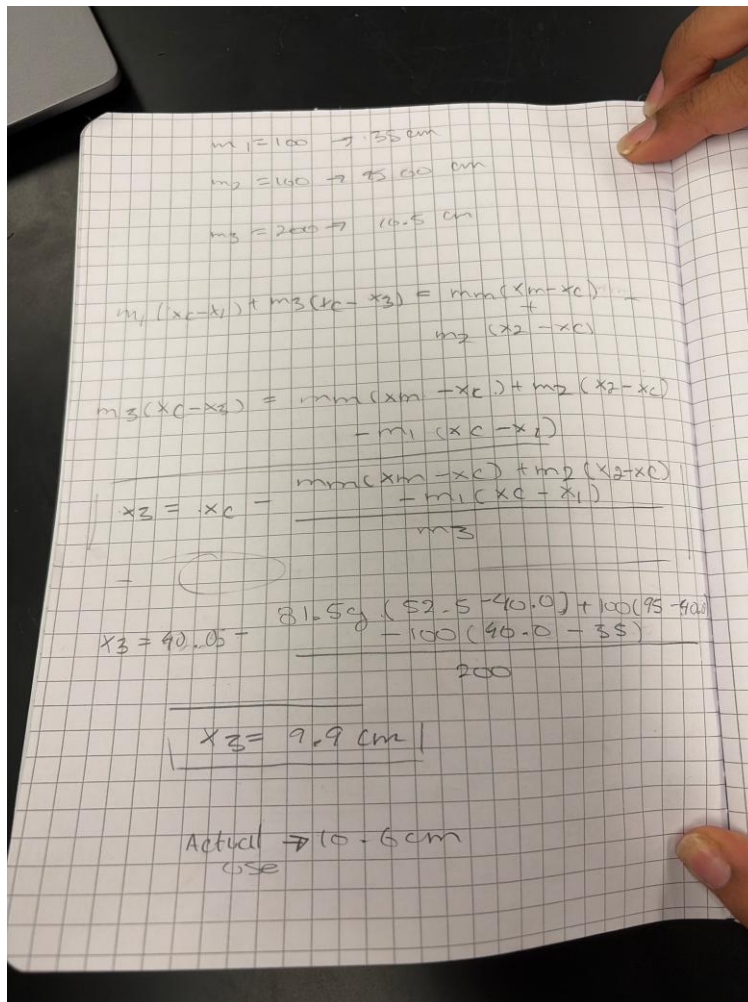
Predict the unknown position

Determine where to hang a total mass of $m_3 = 200$ g in order for the system to be in equilibrium if $m_1 = 100$ g, $x_1 = 35.0$ cm, $m_2 = 100$ g, $x_2 = 95.0$ cm, $x_c = 40.0$ cm.



Position a balancing clamp on the meter stick at the $x_c = 40.0$ cm mark. Suspend a mass $m_1 = 100$ g at $x_1 = 35.0$ cm, $m_2 = 100$ g at $x_2 = 95.0$ cm. Suspend mass $m_3 = 200$ g on the meter stick until the stick is level and in rotational equilibrium. **Record the value for x_3 .**

$x_3 = 10.6$ cm



Calculate the % difference between the predicted x_3 and the experimental value.

$$(10.6\text{cm} - 9.9\text{cm}) / 9.9 \times 100 = 7.07\%$$

Questions

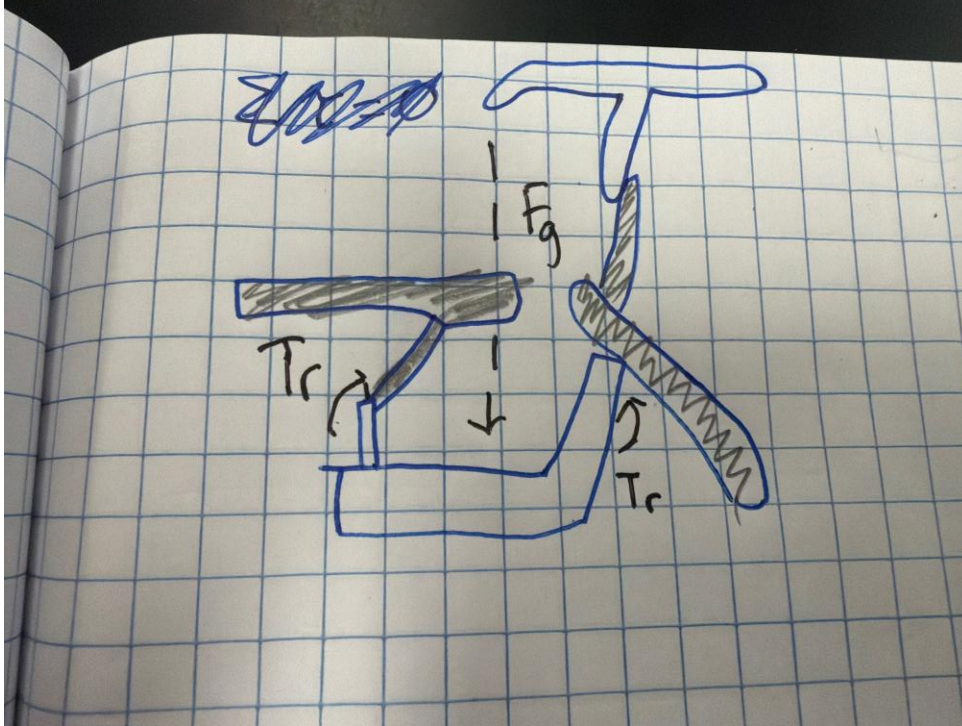
Why is the normal force exerted on the meter pole not considered in each calculation?

Normal force exerted on the meter pole doesn't affect the calculation because it is exerted at the pivot point. This means that no force is added to one arm that isn't added to the other.

Give an example of how what you did in this lab could be applied in the "real-world".

Seesaw with my future child of smaller mass

Draw an extended FBD of each person in this picture. Indicate action/reaction pairs.
Make sure that the sum of the forces and the sum of the torques on each person add up to 0.



Trial	X_c	m_1	x_1	m_2	x_2	m_3	x_3
	1 40cm	386.3 g	30cm	60 g	90 cm	N/A	N/A
	2 40cm		35 cm	100g	95 cm	200g	10.55cm
	3						

avenue/mean

"Stdev"

() Sterrr = Stdev/

m_m	hanger
81.5g	